

Rama Mallavarapu

Software Engineer — ADAS & Autonomous Systems | Sensor Fusion | C++ | ROS

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SUMMARY

Systems and software engineer with 6+ years of experience in ADAS and autonomous driving technology at CARIAD and ZF Mobility Solutions. Specializes in sensor fusion, perception systems, and grid-based localization, with hands-on delivery across prototyping and series development. Proficient in C++, Python, ROS, and MATLAB, working within ASPICE and ISO 26262 safety-critical development environments. Currently transitioning from systems/development engineering into applied ML engineering for simulation and perception.

SKILLS

- Languages & Frameworks: C++, Python, MATLAB, ROS, HTML, Angular, C#
- Domains: Sensor Fusion, Computer Vision, Perception Systems, Grid-Based Localization, Algorithm Development
- Tools: Git, Codebeamer, JIRA, Confluence, Jenkins, CI/CD, GoogleTest, TPT, Conan, Unity
- Standards & Methods: ASPICE (SWE1-SWE6), ISO 26262, ASIL, V-Model, SAFe, Scrum, Design Patterns, Clean Code

PROFESSIONAL EXPERIENCE

Simulation / ML Engineer — CARIAD

Oct 2025 - Present

- Develop and validate simulation-based ML approaches for perception and sensor-fusion performance evaluation.
- Apply ML techniques to strengthen detection quality and robustness KPI frameworks used in series validation.

Systems / Development Engineer — CARIAD

Sep 2022 - Oct 2025

- Validated ADAS systems against the V-Model, including fault analysis and feedback through to series release sign-off.
- Defined perception requirements and interfaces in Codebeamer in compliance with ASPICE.
- Built KPI frameworks for detection quality and robustness, and automated KPI test pipelines for early defect discovery.
- Implemented safety-relevant algorithms in C++ in line with ISO 26262 and ASIL requirements.
- Migrated test cases from TPT to GoogleTest and executed unit, integration, and system testing.
- Led an internal learning group promoting agile and engineering best practices.

Algorithm Developer — ZF Mobility Solutions GmbH

Feb 2018 - Jul 2022

- Developed grid-based sensor fusion algorithms combining radar and camera data for autonomous driving functions.
- Defined KPIs and implemented automated test suites for perception system evaluation.
- Led technical assessment of sensor sets and system architecture for L4 autonomous driving programs.
- Optimized assessment processes and built internal collaboration tooling using Angular, Flask, and HTML.
- Supported integration testing for perception systems.

Master Thesis Student — IEE

Apr 2017 - Dec 2017

- Completed master's thesis research within an automotive sensing environment.

EDUCATION

M.Sc., Automotive Engineering Technology — TH Ingolstadt

2015 - 2017

- Focus: Driver Assistance Systems, Sensing, Mechatronics

B.Tech., Mechanical Engineering — JNTUH College of Engineering, Hyderabad

2010 - 2014

CERTIFICATIONS

- Speexx German – CEFR Level B2.2
- Certificate of Completion: C++ Course
- Automotive SPICE Basics
- Requirements Writing

LANGUAGES

English (Professional Working) | German (Professional Working, CEFR B2) | Hindi (Native) | Telugu (Native)